

# CMPT 354 Mini Project

## Step 1: Project Specifications

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### 1 Project Overview

The goal of this project is to build a database application for a community library. The database will keep one consistent record of the library's collection, loans, fines, events, rooms, members, personnel, donations, volunteers, help requests, and items that may be acquired in the future. The system should help library users find and use services while allowing personnel to maintain accurate records and enforce library rules.

For this project, the library is treated as one location with several social rooms. A person may visit the library without an account, but an active member account is required to borrow an item, register for an event, donate an item, volunteer, or submit a help request through the application.

### 2 Users of the System

The system will support the following users:

- **Visitors** may search the collection and view public event information.
- **Members** may perform all visitor actions. They may also borrow and return items, view their loans and fines, register for events, offer donations, apply to volunteer, and ask a librarian for help.
- **Library personnel** may add and update collection records, process loans and returns, record damaged or lost items, manage events and registrations, review donations and proposed items, approve volunteers, and respond to help requests.
- **Administrators** may perform personnel actions and maintain employee records, library policies, and user access permissions.

Each user will be allowed to view or change only the information required for their role. For example, ordinary members must not be able to view private personnel records or change fines.

## **3 Information Stored by the Database**

### **3.1 Members**

For each member, the library will store a unique membership number, full name, email address, phone number, mailing address, date joined, membership status, and membership expiry date. A membership may be active, suspended, expired, or closed. The database will also preserve the member's loan, fine, event, donation, volunteer, and help-request history.

Each member email address is required and must be unique among member records.

### **3.2 Personnel**

For each employee, the library will store a unique employee number, name, contact information, job title, hire date, employment status, and supervisor when applicable. Job titles may include librarian, event coordinator, cataloguer, and administrator. Only authorized administrators may view or update private personnel information.

Each employee email address is required and must be unique among employee records.

### **3.3 Library Collection**

The collection may contain print books, online books, magazines, scientific journals, music records, films, and other materials. The database will separate the description of a title from the individual copies owned or licensed by the library. This is necessary because several copies of the same title may exist, and each copy may have a different location, condition, or availability.

For every title, the library will store a unique catalogue number, title, material type, creator or author, publisher, publication year, language, subject, description, and a standard identifier such as an ISBN or ISSN when one exists. A standard identifier must be unique when it is present. Each creator will have a unique creator number and a name. A title may have several creators, and one creator may be connected to several titles. Individual issues of magazines and journals will also record their volume, issue number, and publication date; the combination of catalogue number, volume, and issue number uniquely identifies an issue.

Every lendable unit will have a unique internal unit number and a current status. Its status may be available, on loan, reserved for processing, damaged, lost, or withdrawn. A physical copy will also have a unique barcode, shelf location, acquisition date, and condition. A digital licence will instead have a unique licence number, access link, and licence limit. The number of simultaneous digital loans must never exceed the licence limit.

### **3.4 Loans, Returns, and Fines**

For each loan, the database will store a unique loan number, the member, the borrowed lendable unit, the employee who processed the transaction when applicable, the date and time borrowed, the due date, and the actual return date. The return date remains empty while the loan is active.

The normal loan periods are 21 days for print books, 14 days for online books, and 7 days for magazines, journal issues, records, and films. An active member may have at most five active loans. A physical copy cannot be included in more than one active loan at the same time. A member with an overdue item or at least \$20 in unpaid fines cannot begin another loan. Digital access ends automatically on its due date.

If a physical item is returned late, the member is charged \$0.25 for each day after the due date. For each fine, the database will store a unique fine number, the amount, date assessed, reason, payment status, payment date, and the loan that caused the fine. Fine amounts cannot be negative. Lost or seriously damaged items may receive an additional replacement charge recorded with the same loan.

### **3.5 Rooms and Events**

For each social room, the library will store a unique room number, room name, floor, capacity, accessibility information, and availability status. Events may include book clubs, author talks, workshops, art shows, and film screenings.

For each event, the database will store a unique event number, title, event type, description, start time, end time, room, maximum attendance, organizing employee, and event status. An event may be scheduled, cancelled, or completed. All events are free. The maximum attendance cannot be greater than the capacity of the assigned room, and two events cannot use the same room during overlapping times.

Every event will be recommended for one or more audiences, such as children, teens, adults, seniors, families, beginners, or researchers. An audience may be recommended for many events. The recommendation is informational and does not prevent other members from attending unless an event has a stated age or safety restriction.

For each event registration, the system will record a unique registration number, the member, event, registration time, registration status, and whether the member attended. A member may register for the same event only once. Registration is allowed only while the event is scheduled and has remaining capacity.

### **3.6 Donations and Possible Future Items**

The library will keep records of materials that members offer to donate and materials that personnel may purchase in the future. Each proposed item will include a unique proposal number, title, creator, material type, publication information, source of the proposal, date submitted, condition when applicable, notes, and current decision status. The source may be a member donation, a member request, or a personnel purchase suggestion.

A proposal may be pending, under review, accepted, or rejected. When a member submits a donation, the member will be recorded as the donor. A reviewing employee and decision date will be recorded after a decision is made. Accepting a proposal does not immediately make it borrowable; personnel must first create the catalogue and copy records and mark the copy as available. Rejected proposals remain in the history so that the same item is not reviewed repeatedly without a reason.

### **3.7 Volunteering and Help Requests**

A member may submit a volunteer application for a particular event or for general library work. The application will record a unique application number, the member, application date, area of interest, availability, relevant skills, reviewing employee, decision, assignment notes, and optional event. An application may be pending, approved, rejected, or completed. The reviewing employee, assignment notes, and event may remain empty until applicable. Approval must be recorded by an employee before the member is given a volunteer assignment.

A member may also ask a librarian for help. Each help request will record a unique request number, member, request category, question or description, submission time, priority, status, assigned librarian, response, and completion time. The assigned librarian, response, and completion time may remain empty until applicable. A request may be open, assigned, resolved, or closed. A request cannot be marked resolved without a response and a completion time.

## 4 Required Application Operations

The completed application must allow an authorized user to perform the following operations while preserving all library rules:

1. Search for an item by title, creator, subject, material type, catalogue number, ISBN/ISSN, or availability, and view the location or online access information that the user is permitted to see.
2. Borrow an available item after checking the member's status, current number of loans, overdue items, unpaid fines, and the availability of the physical copy or digital licence.
3. Return an active loan, update the copy's condition and availability, and assess a fine when the return is late, lost, or damaged.
4. Submit and review a possible donation, and convert an accepted donation into catalogue and copy records after processing.
5. Search for events by title, type, date, audience, or availability.
6. Register a member for a free event after checking event status, room capacity, and duplicate registration.
7. Submit and review a volunteer application and, when approved, record its assignment or related event.
8. Submit, assign, answer, and close a request for help from a librarian.

Personnel must also be able to view operational reports, including currently borrowed and overdue items, unpaid fines, item availability, room schedules, event registrations and attendance, pending donations, volunteer applications, and unresolved help requests.

## 5 Data Integrity and Record-Keeping Rules

The following rules apply throughout the database:

- Membership numbers, employee numbers, creator numbers, catalogue numbers, lendable-unit numbers, physical-copy barcodes, digital-licence numbers, room numbers, loan numbers, fine numbers, event numbers, registration numbers, proposal numbers, volunteer-application numbers, and help-request numbers must be unique. Member and employee email addresses are also unique within their respective record types. A periodical issue is unique by its catalogue number, volume, and issue number, and a member-event pair is unique in event registrations.

- Required names, titles, dates, and status values cannot be empty, and all stored dates must follow a consistent format.
- A due date must be after its borrowing date, an event's end time must be after its start time, and a payment or completion date cannot be earlier than the record that caused it.
- A loan must refer to an existing member and lendable unit. A fine must refer to an existing loan. An event must refer to an existing room and organizing employee. Registrations, item proposals, volunteer applications, and help requests must refer to their related existing records.
- Status values must come from the allowed values described above. Counts, capacities, licence limits, and monetary amounts cannot be negative.
- Important history such as completed loans, paid fines, past events, reviewed proposals, and resolved requests will not be deleted during normal operation. If a record is no longer active, its status will be changed so that reports remain accurate.
- The responsible employee will be recorded for employee-processed loans, event organization, proposal review, volunteer review, and help-request assignment whenever the corresponding workflow applies.

## 6 Scope Limitations

The first version of the system will not handle employee payroll, paid event tickets, inter-library loans, room rentals, or the purchase of commercial digital licences. It will only keep the information needed to support the library services and application operations described above.